

Functions and Limits Page 1

A function maps each input to exactly one output: $f(x)$.

Domain = allowed inputs. Range = possible outputs.

Vertical line test: crosses once $\Rightarrow$ it is a function.

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A limit describes the value a function approaches.

Notation: $\lim_{x \rightarrow a} f(x) = L$

Limits can exist even when $f(a)$ is undefined.

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Evaluating limits:

1. Try direct substitution.
2. Factor and cancel for $0/0$.
3. Use conjugates for radicals.